

CE480- HW6

Breakwater Layout Design Report

Group 3

Hamidreza Khalaj Zahraei (300204093)

Ali Özuysal (310204059)

Coastal Engineering

This report currently uses example values. Replace `USER_INPUTS` in the Python file with the values from HW5 before submitting.

Objective

Design/check the breakwater layout so the diffracted wave height at the quay wall is below the serviceability limit required for the selected harbor type.

Homework 6 Workflow

1. Take the 10-year design wave height H_i and corresponding period T from HW5.
2. Use the wave angle α from HW5, or calculate it by refraction analysis.
3. Select the correct diffraction diagram using the angle between the incident wave and the breakwater.
4. Read the diffraction coefficient K_d at the quay-wall location.
5. Check $H_q = K_d H_i$ against the allowable harbor limit.

Current Design Check Summary

Design item	Value
Incident design wave height, H_i	1.200 m
Diagram diffraction coefficient, K_d	0.220
Calculated quay-wall wave height, $H_q = K_d H_i$	0.264 m
Allowable wave height, H_{allow}	0.300 m
Decision	ACCEPTABLE

This summary is automatically updated when the values in `USER_INPUTS` are changed and the Python file is run again.

Design equation chain

HW5 wave condition $\rightarrow$ wavelength $L \rightarrow$ diagram coordinates r/L and $\theta \rightarrow$ diffraction coefficient $K_d \rightarrow$ quay-wall height $H_q = K_d H_i \rightarrow$ final check $H_q \leq H_{allow}$. This is the complete logic used throughout the report.

Design Basis

What is being checked and why

The breakwater layout is checked as a serviceability problem. The main question is not whether the structure survives the offshore design wave; the question is whether the wave agitation reaching the quay wall is small enough for safe and practical harbor operation.

The incident design condition is taken from HW5: the 10-year design wave height H_i in front of the breakwater and its corresponding period T . Those values define the wave length scale, the diagram coordinates, and the wave height that must be reduced by diffraction.

Wave diffraction occurs because the breakwater blocks part of the incoming wave front. Near the breakwater head, wave energy bends into the protected area. The farther the quay wall is inside the shadow zone, and the less directly exposed it is to the opening, the smaller the diffraction coefficient K_d should become.

For a marina layout, the assignment limit used here is $H_{allow} = 0.30$ m. The design passes only when the calculated quay-wall wave height H_q is less than or equal to that limit.

Important assumptions

1. The design wave height, period, and approach angle are taken as already established by HW5.
2. The selected diffraction diagram represents the breakwater-head geometry well enough for homework-level design.
3. The quay-wall check point represents the most critical protected location.
4. Reflection, resonance, transmission through the structure, and local berth geometry are not included unless the instructor asks for them.
5. If the diagram value gives H_q above the allowable limit, the layout must be changed and checked again.

Calculation Method

Formulas are written where they are used

1. Convert the HW5 wave period into a wavelength

The design wave period T is needed because diffraction diagrams are read using the non-dimensional distance r/L . First, the deep-water estimate is $L_0 = g T^2 / (2 \pi)$. Because the harbor has finite depth, the wavelength used in the layout check is corrected with the dispersion equation $L = (g T^2 / 2 \pi) \tanh(2 \pi h / L)$. The script solves this equation iteratively until L stops changing.

2. Locate the quay-wall point on the diffraction diagram

The quay-wall check point is measured from the breakwater head. If the breakwater tip is (x_t, y_t) and the quay point is (x_q, y_q) , the distance is $r = \sqrt{(x_q - x_t)^2 + (y_q - y_t)^2}$. The value used on the diagram is not r alone, but r/L . The angular location is also needed, so the script uses $\theta = \text{atan2}(y_q - y_t, x_q - x_t) - \beta_{BW}$, where β_{BW} is the breakwater axis direction.

3. Read the diffraction coefficient

The correct diagram is selected from the incident wave angle relative to the breakwater. At the computed r/L and θ , the diagram gives the diffraction coefficient K_d . By definition, $K_d = H_q / H_i$, so it represents the fraction of the incident design wave height that reaches the quay wall.

4. Calculate the wave height at the quay wall

After K_d is read from the diagram, the quay-wall wave height is calculated directly as $H_q = K_d H_i$. This is the main homework result because it tells how much wave agitation remains after the breakwater shelters the harbor.

5. Check the allowable harbor limit

The design passes when $H_q \leq H_{allow}$. The assignment gives $H_{allow} = 0.30$ m for marinas and fishery harbors, and $H_{allow} = 0.50$ m for container terminals. The required coefficient can also be written as $K_{d,req} = H_{allow} / H_i$. Therefore, the layout is acceptable only if the diagram value satisfies $K_d \leq K_{d,req}$.

6. Report the safety margin

The safety margin is $\text{Margin} = H_{allow} - H_q$. A positive value means the quay-wall wave height is below the limit. A negative value means the breakwater layout must be revised and the diagram must be read again.

Worked use with the current input values

For the present example, $T = 6.50$ s and $h = 8.00$ m give $L = 50.248$ m. The distance from the breakwater tip to the quay wall is $r = 107.005$ m, so the diagram coordinate is $r/L = 2.130$. With $K_d = 0.220$, the protected wave height is $H_q = K_d H_i = 0.220 \times 1.200 = 0.264$ m. The required value is $K_{d,req} = H_{allow}/H_i = 0.250$.

Wave Theory Notes

Why wavelength matters in the diffraction diagram

Diffraction diagrams normally use non-dimensional coordinates. That means the actual plan distance from the breakwater tip to the quay wall is divided by the local wavelength. A 100 m distance is not equally protected for every wave; it is more protected for short waves than for long waves.

Wave parameter	Value
Deep-water wavelength L_0	65.965 m
Finite-depth wavelength L	50.248 m
Water depth h	8.000 m
Relative depth h/L	0.159
Depth classification	intermediate-water range
Wave celerity C	7.731 m/s

For this homework, the wavelength is mainly used to compute r/L before reading the diffraction diagram. If the water depth or wave period changes, L changes, so the same physical layout can correspond to a different point on the diagram.

Input Data

Replace example values with the HW5/project values.

Input	Value
Project	Breakwater Layout Design Report
Harbor type	marina
Design wave height H_i	1.200 m
Design wave period T	6.500 s
Water depth h	8.000 m
Incident angle to breakwater	35.00 deg
Diffraction coefficient K_d	0.220
Breakwater axis	0.00 deg
Breakwater tip	(0.0, 0.0) m
Quay-wall check point	(85.0, -65.0) m

Calculation Results

Diffraction check at the quay-wall point.

Quantity	Result
Wavelength L	50.248 m
Wave number k	0.12504 rad/m
Wave celerity C	7.731 m/s
Tip-to-quay distance r	107.005 m
Dimensionless distance r/L	2.130
Quay angle from breakwater	-37.41 deg
Wave height at quay H _q	0.264 m
Allowable wave height	0.300 m
Required K _d or lower	0.250
Safety margin	+0.036 m
Decision	ACCEPTABLE

The safety margin is positive, so the checked point satisfies the selected harbor limit.

Calculation sequence

1. The finite-depth dispersion equation gives $L = 50.248$ m for $T = 6.50$ s and $h = 8.00$ m.
2. The geometry gives $r = 107.005$ m, therefore $r/L = 2.130$ for the diffraction diagram.
3. The selected diagram gives $K_d = 0.220$ at the quay-wall point.
4. The protected wave height is $H_q = K_d H_i = 0.220 \times 1.200 = 0.264$ m.
5. The margin is $H_{allow} - H_q = 0.300 - 0.264 = +0.036$ m.

Result Interpretation

Engineering meaning of the calculated values

With the current input values, the diagram coefficient is $K_d = 0.220$. The calculated quay-wall wave height is $H_q = 0.264$ m, while the allowable limit is $H_{allow} = 0.300$ m. The resulting safety margin is $+0.036$ m, so the decision is: ACCEPTABLE.

What the numbers mean

The required coefficient is $K_{d,req} = 0.250$. Any layout and diagram reading with K_d less than or equal to 0.250 satisfies the wave-height criterion for the chosen harbor type. A lower K_d means stronger sheltering. A higher K_d means the quay wall receives too much wave energy.

How to improve the design if necessary

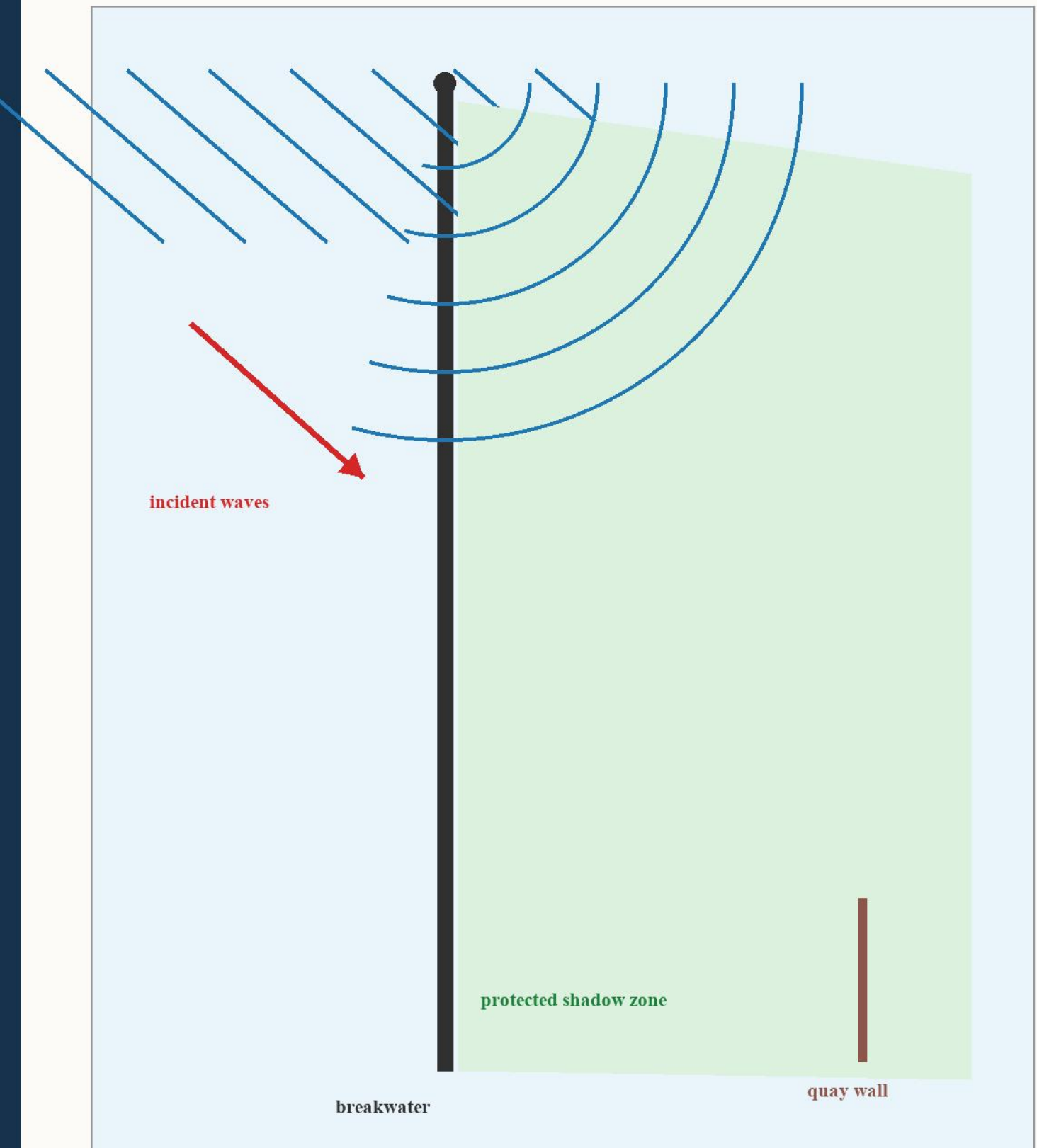
The most direct way to improve the result is to move the quay wall farther inside the protected region or increase the breakwater projection so the critical quay point lies on a lower K_d contour. Rotating the breakwater or narrowing the entrance can also reduce direct wave penetration. Each layout change should be followed by a new diagram reading, because the value of K_d depends on both angle and non-dimensional position.

Submission note

Before submission, replace the example inputs with your HW5/project values and attach or cite the diffraction diagram used to read K_d if your instructor requires showing the graphical reading.

Conceptual Diffraction Figure

Wave bending around the breakwater head



This conceptual figure shows the physical reason for using K_d . The breakwater blocks the direct wave front, but energy bends around the head. The diagram coefficient K_d estimates how much of the incident wave height reaches the selected quay-wall point.

Diffraction Diagram Use

How the diagram value enters the design

Use the incident wave angle relative to the breakwater to choose the correct diffraction diagram. Then locate the quay-wall point using the dimensionless radius r/L and its angular position behind the breakwater. The diagram gives K_d at that point.

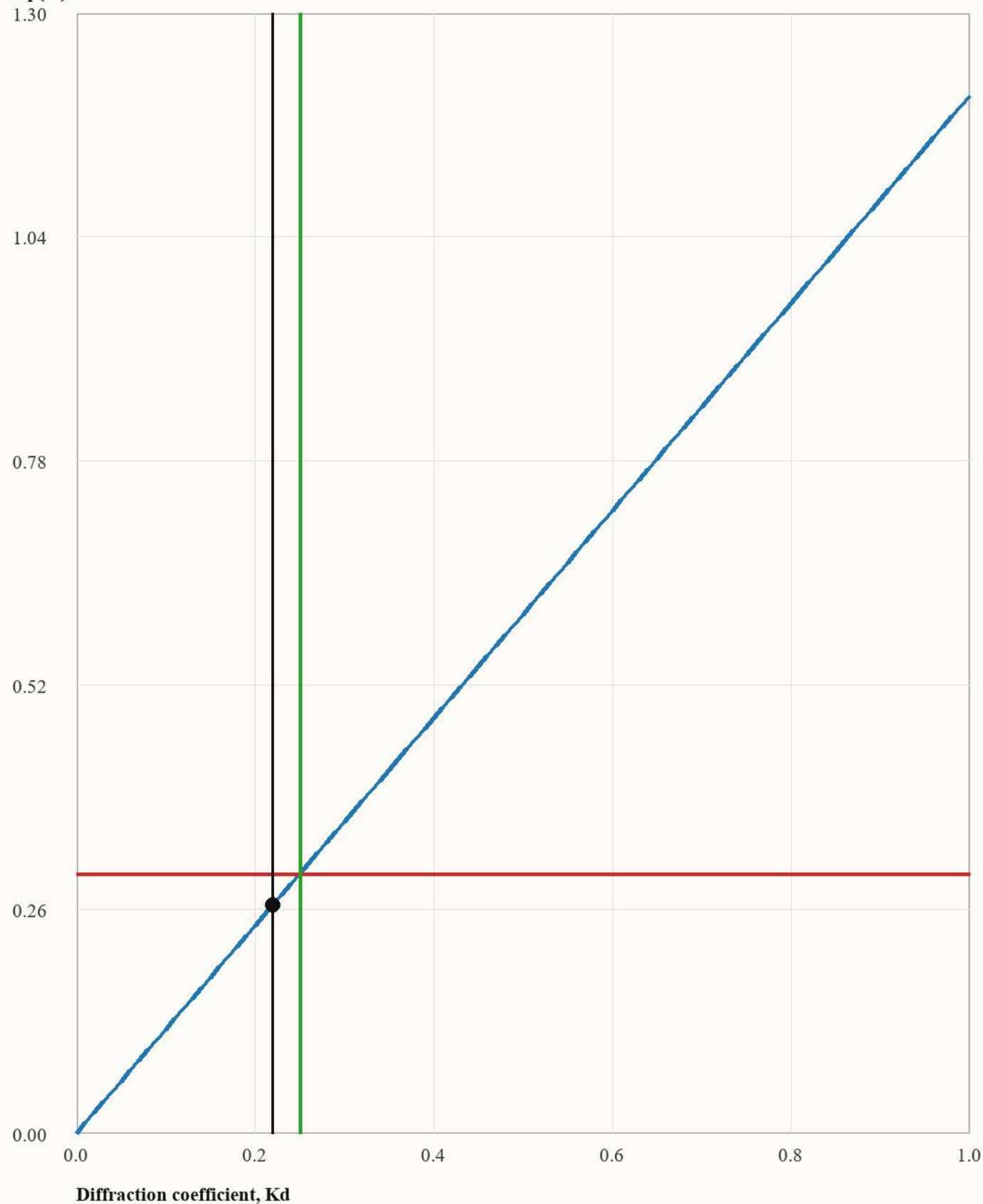
Step	Action
1	Calculate wavelength L from T and h .
2	Measure r from the breakwater tip to the quay-wall check point.
3	Calculate r/L .
4	Measure angular position of the quay point behind the breakwater.
5	Read K_d from the diagram chosen for the incident angle.
6	Compute $H_q = K_d H_i$ and compare with the allowable limit.

Important: this script does not replace the required diffraction diagram. It records the diagram coefficient and completes the design check consistently.

Wave Height Sensitivity

Hq as a function of the diffraction coefficient

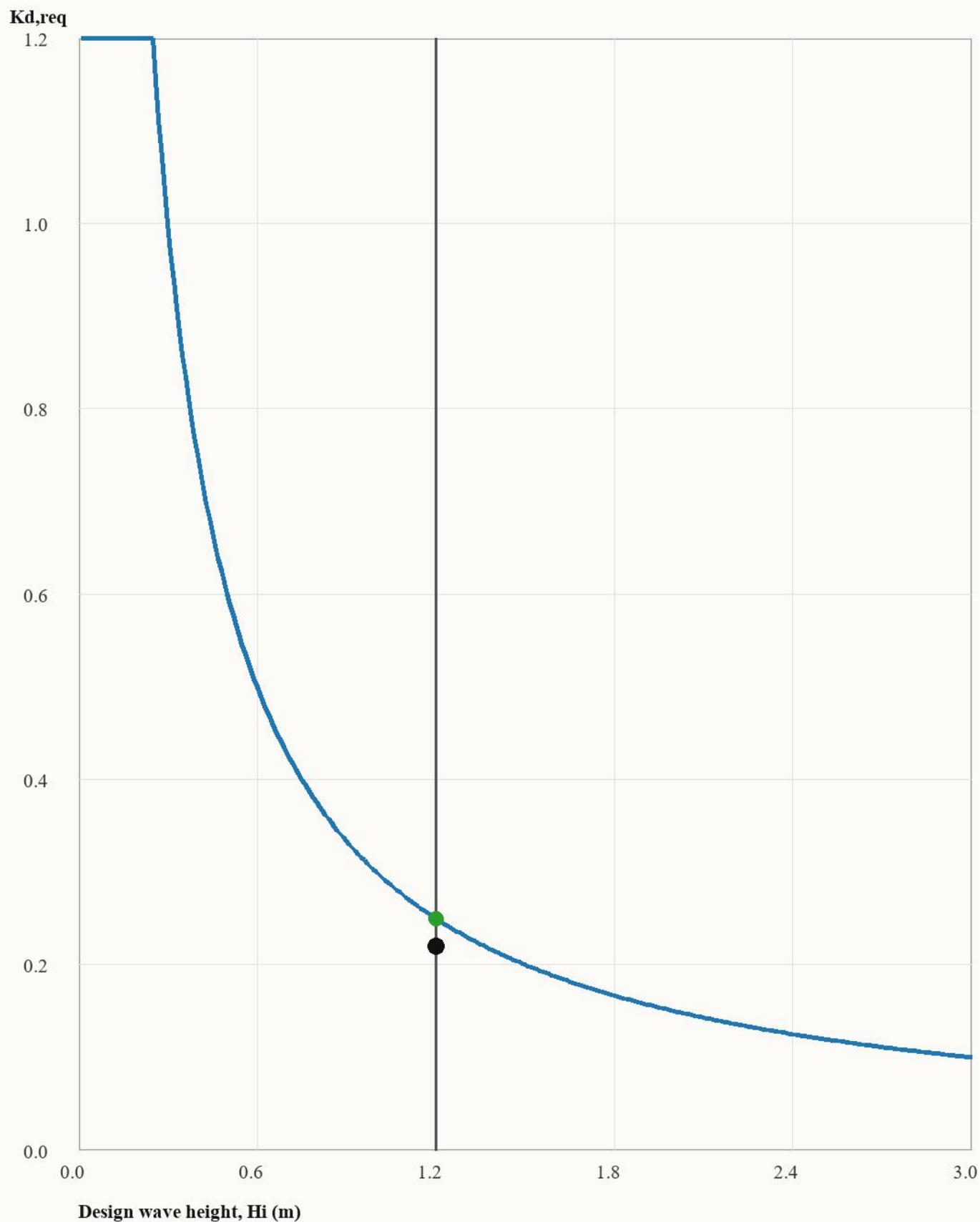
Hq (m)



To satisfy the criterion, the final layout must give $K_d \leq 0.250$. If the diagram value is larger than this, revise the layout and read a new diffraction coefficient.

Required Diffraction Reduction

Maximum Kd allowed for different design wave heights



Design wave height, H_i (m)

Blue curve: $K_{d,req} = H_{allow} / H_i$

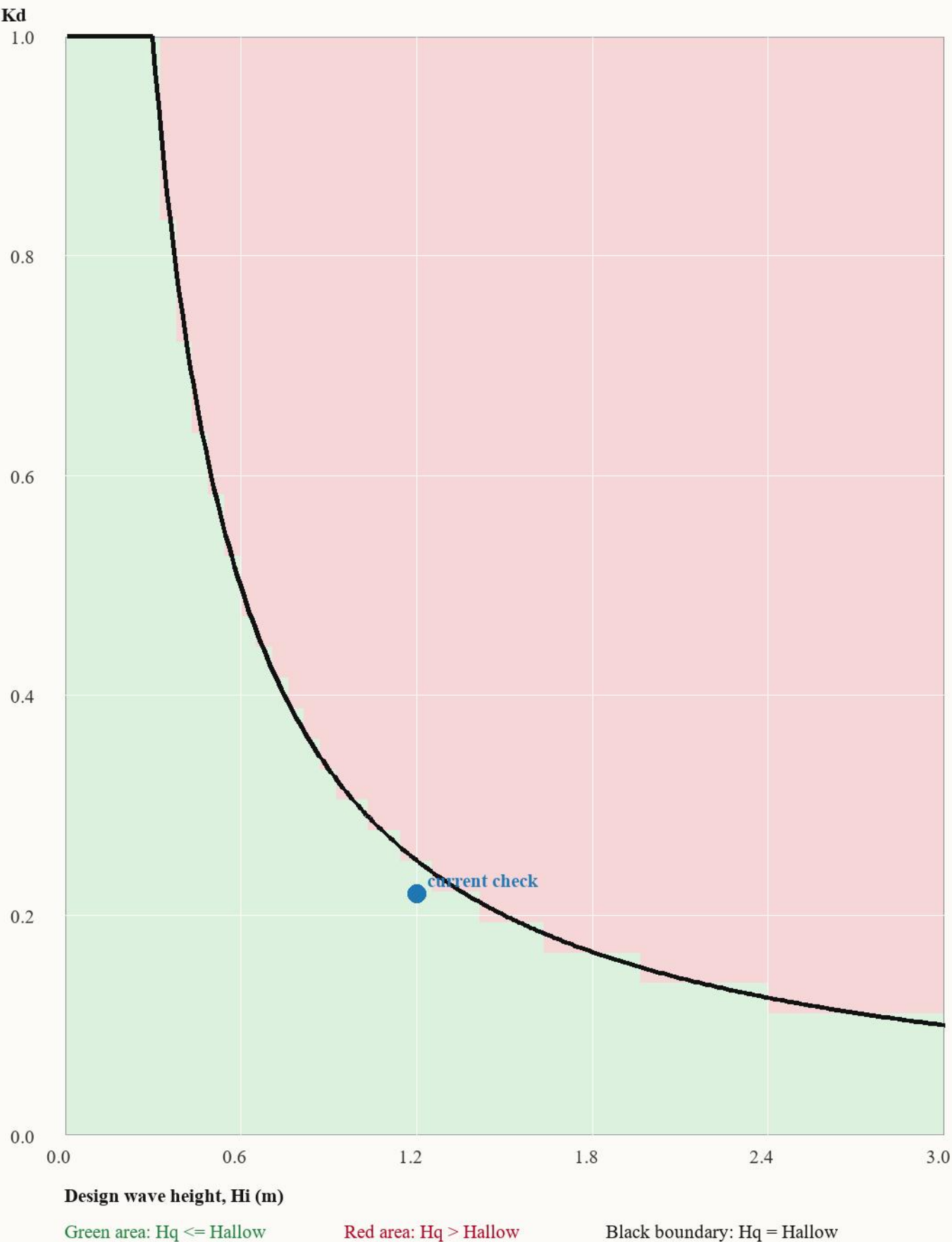
Black point: diagram K_d at current H_i

Green point: required K_d

This figure shows why larger design waves require stronger sheltering. If H_i increases while the allowable harbor agitation stays fixed, the maximum acceptable K_d becomes smaller.

Pass/Fail Design Envelope

Combination of H_i and K_d that satisfies the harbor limit



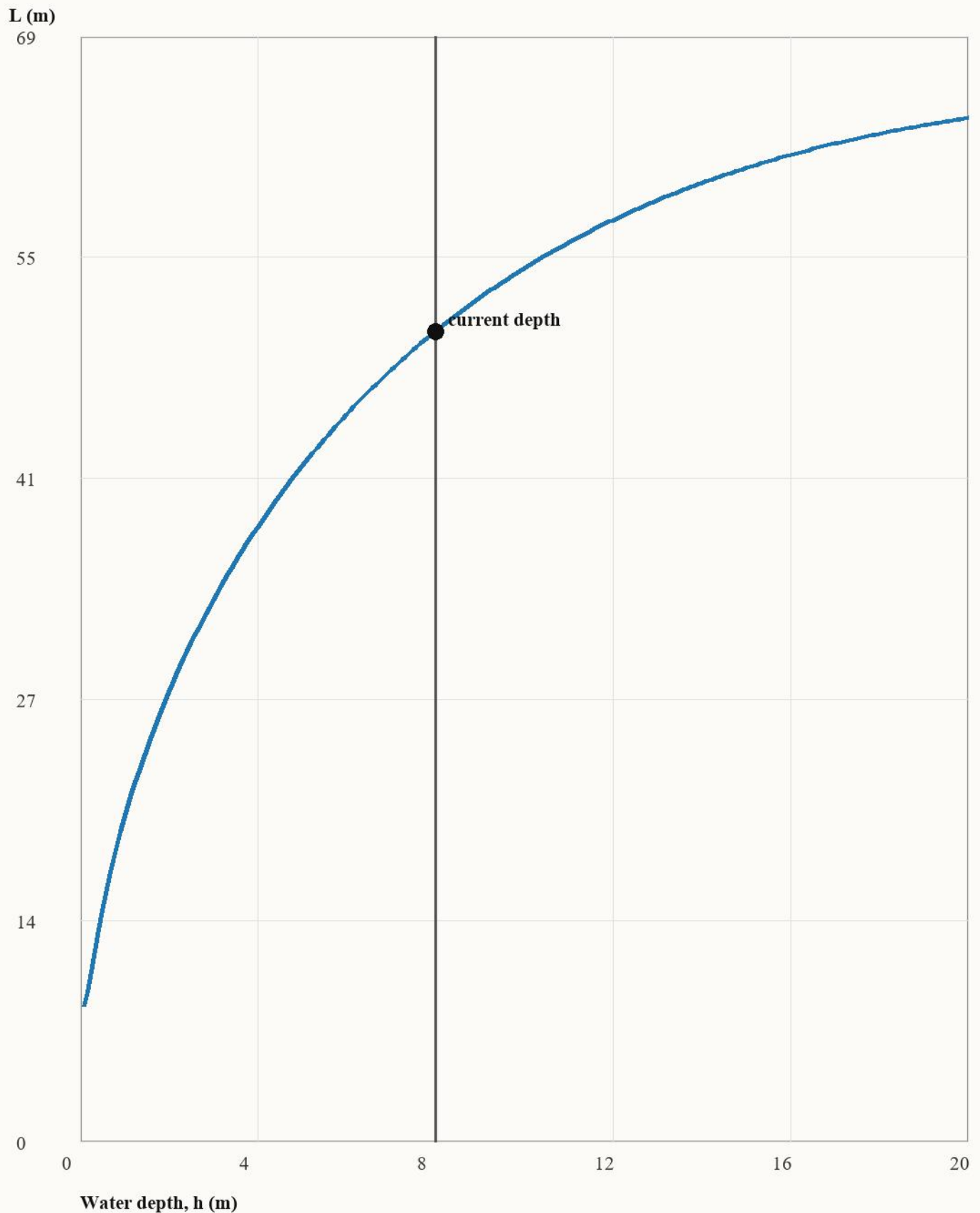
The layout passes only inside the green region. Moving downward means reducing K_d by improving the layout. Moving left means the design wave height is smaller, which is not a layout change and should only come from the HW5 design condition.

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Wavelength Versus Depth

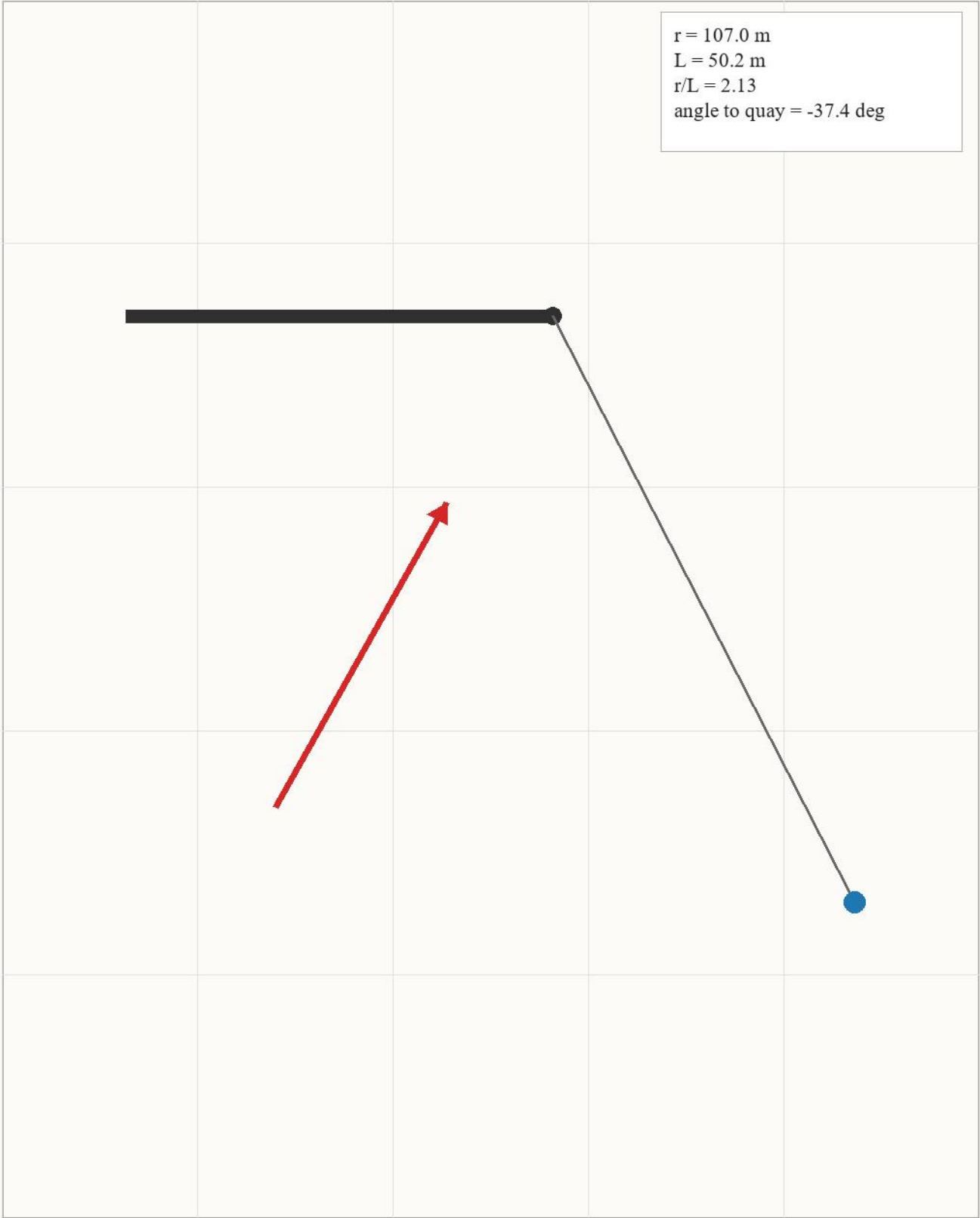
Effect of water depth for the selected wave period



This figure explains why the depth used in the layout check matters. A different depth gives a different wavelength, which changes r/L and therefore the point read from the diffraction diagram.

Plan Schematic

Local layout coordinates used for the check



Black line: breakwater | Blue point: quay-wall check point | Red arrow: incident waves | Gray line: r from tip to quay

If the Layout Fails

Revision rule

For this input set, the layout must satisfy $K_d \leq 0.250$. If the diffraction diagram gives a larger value, the wave height at the quay wall is too large.

Possible layout revisions

1. Increase the breakwater length so the quay wall is deeper inside the shadow zone.
2. Rotate or reposition the breakwater so incident waves have less direct access to the quay.
3. Reduce or reorient the harbor entrance opening.
4. Add a secondary or return breakwater if a single breakwater cannot satisfy the limit.
5. Move the quay-wall check point or layout farther from the diffraction tip, if the site plan allows it.

After each revision, select/read the new diffraction coefficient from the proper diagram and repeat the check.

Revision effect on the calculation

Revision	Expected calculation effect
Longer breakwater	Usually moves the quay point deeper into lower K_d contours.
Different orientation	Changes the incident angle and may require a different diagram.
Narrower entrance	Reduces direct exposure and can lower the selected K_d value.
Secondary breakwater	Creates another sheltering boundary when one arm is not enough.
Relocated quay	Changes r and θ , so the diagram reading changes.

The revised design should be accepted only after the new diagram reading gives $H_q = K_d H_i$ less than or equal to the allowable wave height.